

Roll No. :

MID SEMESTER EXAMINATION MARCH 2025

Name of the Program: **B.Tech**

Semester: 8th

Name of the Course : Agile Software Engineering

Course Code: TCS 855

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	
(a)	Your development team is working on a new mobile application. Midway through the project, the client requests several major changes to the application's core features. How would Scrum and XP handle these changes differently? Which approach provides more flexibility in this scenario, and why?	CO1
	OR	
(b)	A software company needs to release specific features as quickly as possible while ensuring long-term maintainability. Some team members argue that FDD's feature-based approach would work best, while others advocate for Scrum's iterative model. How does the focus on features (FDD) vs. iterations (Scrum) impact project delivery and flexibility? Which would be a better fit for a fast-moving product?	CO2
Q2	(10 Marks)	
(a)	The company has started using Agile, but some teams are struggling with incomplete sprints, frequent scope creep, and unclear roles. What are the most common pitfalls companies face when transitioning from Waterfall to Agile, and how can they be avoided?	CO1
	OR	
(b)	Compare and contrast the Product Backlog and the Sprint Backlog in terms of ownership, contents, and purpose.	CO3
Q3	(10 Marks)	
(a)	A newly formed Agile team is confused about the roles of the Product Owner, Scrum Master, and Scrum Developer. Explain the responsibilities of each Scrum role and how they contribute to a successful Agile project.	CO1
	OR	
(b)	A product owner insists that the team focus only on new features rather than spending time refactoring code. However, developers believe that the existing codebase has become difficult to maintain due to technical debt. How should the team convince the product owner of the importance of refactoring while still delivering new features? What Agile strategies can help balance both?	CO3
Q4	(10 Marks)	
(a)	A Scrum team is struggling with inconsistent story point estimations. Some team members assign higher points for the same tasks, while others assign lower points, causing confusion in Sprint planning. What strategies can the team use to achieve more consistent and reliable Agile estimations?	CO2

	OR	CO3
(b)	A project manager wants to track the progress of a Scrum team and forecast project completion. They are unfamiliar with Agile velocity and burndown charts but need insights into the team's performance. Explain Agile velocity and how it helps in forecasting project completion. Describe how burndown charts work and their role in measuring a team's performance and Sprint progress.	
Q5	(10 Marks)	
(a)	A Scrum team is reviewing user stories during backlog refinement. Some stories are too large and unclear, making them difficult to estimate and implement. Explain the INVEST criteria (Independent, Negotiable, Valuable, Estimable, Small, Testable) and how it helps in writing better user stories.	CO1
	OR	
(b)	Two developers working in a pair programming setup have different coding styles and often disagree on implementation approaches, slowing down progress. As an XP Coach, how would you help them resolve their differences and work effectively together?	CO2